

BdMO 2026 National Camp Sequences, Series & Ad-hoc Problem Set

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§1 Problems

Problem 1.1. Find the sum

$$\frac{1}{\sqrt{1} + \sqrt{2}} + \frac{1}{\sqrt{2} + \sqrt{3}} + \cdots + \frac{1}{\sqrt{n} + \sqrt{n+1}}.$$

Problem 1.2. Let $\{a_k\}$ be an arithmetic progression with common difference d . Compute

$$\sum_{k=1}^n \frac{1}{a_k a_{k+1} a_{k+2}}$$

Problem 1.3. Suppose that a sequence $a_1, a_2, \dots$ of positive real numbers satisfies

$$a_{k+1} \geq \frac{ka_k}{a_k^2 + (k-1)}$$

for every positive integer k . Prove that $a_1 + a_2 + \cdots + a_n \geq n$ for every $n \geq 2$.

Problem 1.4. Let a be a real number not equal to $0, \pm \frac{1}{\sqrt{3}}$, or $\pm \sqrt{3}$, and define a sequence $x_1, x_2, x_3, \dots$ by

$$x_{n+1} = \frac{\sqrt{3}x_n + 1}{\sqrt{3} - x_n} \quad \text{for } n \geq 1.$$

Prove that this sequence is periodic, that is, there exists a positive integer k such that $x_{n+k} = x_n$ for all positive integers n .

Problem 1.5. A sequence of real numbers $a_0, a_1, \dots$ is defined as follows. a_0 is an arbitrary real number and for $n \geq 0$, $a_{n+1} = \lfloor a_n \rfloor \{a_n\}$. (Here, $\lfloor x \rfloor$ denotes the greatest integer less than or equal to x , and $\{x\} = x - \lfloor x \rfloor$). Prove that $a_n = a_{n+2}$ for n sufficiently large.

Problem 1.6. A sequence of positive real numbers a_n is defined by

$$a_0 = 1, \quad a_1 = 3, \quad a_{n+2} = \frac{a_{n+1}^2 + 2}{a_n}.$$

Show that all terms of the sequence are positive integers.

Problem 1.7. Determine all positive integers M such that the sequence $a_0, a_1, a_2, \dots$ defined by

$$a_0 = M + \frac{1}{2} \quad \text{and} \quad a_{k+1} = a_k \lfloor a_k \rfloor \quad \text{for } k = 0, 1, 2, \dots$$

contains at least one integer term.

Problem 1.8. Find a formula in compact form for the general term of the sequence defined recursively by $x_1 = 1$, $x_n = x_{n-1} + n$ if n is odd, and $x_n = x_{n-1} + n - 1$ if n is even.

Problem 1.9. A sequence $(a_n)_{n \geq 0}$ satisfies $a_{m+n} + a_{m-n} = \frac{1}{2}(a_{2m} + a_{2n})$ for all integers m, n with $m \geq n \geq 0$. Given that $a_1 = 1$, find a_{2003} .

Problem 1.10. Find the general term of the sequence defined by $x_0 = 3, x_1 = 4$ and

$$x_{n+1} = x_{n-1}^2 - nx_n$$

for all $n \in \mathbb{N}$.

Problem 1.11. Prove that

$$\frac{1}{2} \cdot \frac{3}{4} \cdot \dots \cdot \frac{2n-1}{2n} < \frac{1}{\sqrt{3n}}$$

for all positive integers n .

Problem 1.12. A sequence of real numbers $a_1, a_2, \dots$ is called *interesting* if $a_1 > 0$ and, for every $n \geq 2$,

$$a_1 a_2 \dots a_{n-1} a_n = a_1 + a_2 + \dots + a_{n-1}.$$

What is the maximum number of integers that such a sequence can contain?

Note: the product has n factors and the sum has $n - 1$ summands.

Problem 1.13. Fix a positive integer n . Define sequences $a, b, c \in \mathbb{Q}^{n+1}$ by $(a_0, b_0, c_0) = (0, 0, 1)$ and

$$a_k = (n - k + 1) \cdot c_{k-1}, \quad b_k = \binom{n}{k} - c_k - a_k, \quad \text{and} \quad c_k = \frac{b_{k-1}}{k}$$

for each integer $1 \leq k \leq n$.

Determine for which n it happens that $a, b, c \in \mathbb{Z}^{n+1}$.